



*Instructor Information

Get to Know Your Instructor

Hi! My name is Jamie Kneisley. Please address me as **Dr. Kneisley** (pronounced like “nicely” but with a z sound instead of a c sound) or **Dr. K** and be sure to let me know if I mispronounce your name or if you would like to be called something other than what is on the roster. I made a [short video](#) to help you get to know me a little bit.

Contact Information

- Phone Number: (432) 685-4645
- Email: jkneisley@midland.edu
- Office: MHAB 122
- Office Hours: Mondays 12:30 to 2:30 PM, Tuesdays & Thursdays 11 AM to 12:30 PM, Fridays 9:00 to 11:00 AM

Email Etiquette

Preferred Method

Please note that I **prefer Canvas messages** to email through my Midland College account for a variety of reasons.

Before Sending Your Email

- Include a **subject that summarizes your request** or question topic
- If you are emailing about a specific assignment, please **use the name of the assignment** as it appears in Canvas.
- If you are emailing about a technical issue, **screenshots help** me understand exactly what you are describing and often help us avoid going back and forth. [Here](#) is a Chrome extension that will allow you to take screenshots.
- If you choose to email me through my MC (Midland College) address, please be sure to **include your name and course**.

Office Hours

Office hours are times that I have scheduled to meet with and talk to you outside of class time. I have a physical office in **MHAB Room 122**, as well as an online office on **Teams** (see left-hand navigation). During the summer, **it is important that you make an appointment** so that I don't miss you. You may do so by sending me an email.

What Can I Expect from Office Hours?

You may come by if you have course- or content-related questions, if you are struggling with school or life, or even if you just want to visit. Again, I **want you to come by for whatever you feel like** you need! You are *not* wasting my time. The ONLY caveat to this statement is this: office hours are not a time for me to reteach the lesson to you. I will be glad to help you if you have done the work FIRST to try to identify where you are stuck.

So, what does this mean? It means that you use the available resources to try to learn the material. Some things to try:

- Borrow a book from the Math Lab.
- See the topical videos from [reliable online sources](#)
- Ask a classmate to explain it
- .
- Attempt the practice assignments (in MyLab or from the book).

If you are still stuck at this point, please see me. [This article](#) gives a great summary of what to expect that I will expect of you during office hours and [this video](#) describes other ways to get ready for your visit when you need help.

TO DO: Find Dr. Kneisley's physical or online office.



*Course Materials

Required Materials

The following materials are **required** for this class:

- [SCIENTIFIC calculator](#)
- Pencil/writing materials
- Paper
- Printer (free in the MC FLRC) **OR** tablet & stylus and the ability to import and annotate PDFs.
- Access to a scanner **OR** scanner app (available in the MC Math Lab or FLRC)
- Computer/laptop (you may use the computers in the MC Math Lab or the FLRC)
- Textbook access (can be purchased through the [MC Bookstore](#) or directly from Pearson through our Canvas course)
- Internet access (available all over MC and in most public libraries)

TO DO: Access your materials in MyMath Lab.

Suggested Materials

The following materials are **suggested** but not required for this class:

- Graph paper
- Ruler
- Colored pencils or fine-tipped markers
- Desmos account or access

Calculators

It is common that students bring calculators to class that are not allowed on assessments. You will not be prevented from using these calculators during class or on practice assignments, but you will be asked to put them away when working on proctored assessments. Below are non-exhaustive lists of allowed and disallowed calculators.

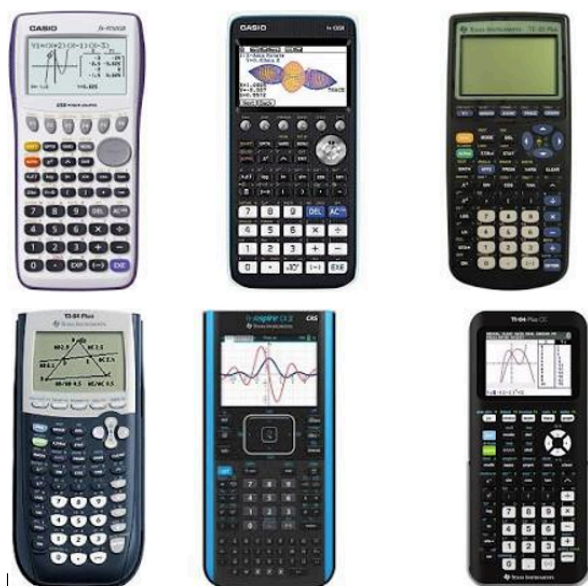
Allowed But Insufficient Calculators



Allowed Calculators



Not Allowed Calculators



TO DO: If you have not used your calculator for a previous course at MC, send Dr. Kneisley a picture of your calculator to verify that it is allowed and sufficient for the course.



Course Navigation

Course Navigation

To ensure you are seeing all the resources available and progressing through the course as intended, I strongly recommend you view the course ***in a web browser on a computer***. Some items, such as the Canvas grade book, do not display properly in the Canvas app. I suggest you add a shortcut to the Canvas webpage directly to the home screen on [your computer](#) and [mobile device](#) rather than using the Canvas app. (Note: Canvas does not always play well with Safari. Your best bet is to use Chrome.)

Course Homepage

Whether you are using a browser on a computer or mobile device or the Canvas Student app, I recommend that you use the Canvas course homepage to navigate the course. From this page, you can get to the weekly overview and the lesson materials, as well as helpful resources for a variety of needs, the Ask a Peer|Help a Peer Discussion, and the written assignments for the course.

TO DO: Find the Course Homepage in the internet browser on a computer and on a phone, then explore the page to familiarize yourself with what is there.

Weekly Announcements

In addition to suggestions for success in this course and important course information, the weekly announcements also offer access to the weekly overview which tells you what to expect and how you should proceed. Instructions for customizing how you receive notifications for announcements and other activities in Canvas can be viewed [here](#).

TO DO: Customize your Notification settings to receive immediate notification of announcements, at least for this course.

Module & Lesson Requirements

You should navigate through Canvas from the weekly announcements or the Canvas course homepage and using the next buttons at the bottom of the screen.

NOTE: ALL items in this BOSS module must be completed to unlock the rest of the course. There are also module requirements for each subsequent module in the course to prevent you from missing out on important assignments/resources.

Each lesson in the course can only be opened by completing the previous modules, ***including the Midterm and Final Exam modules.*** Some items in the module merely need to be viewed, submitted, or marked as done, and others require a minimum score.

[Viewing Module Requirements as a student:](#)

▼ Beginning of Semester		Complete All Items	⊖	Module incomplete
Getting Started				
	Start Here: Welcome to College Algebra!	Jul 7 Marked done	✓	Item complete
	Student Profile	Jul 7 1 pts Score at least 1.0	⊖	Item incomplete
	Flipgrid Introductions	Jul 7 1 pts Mark done	○	Item not yet attempted

In a Browser

If you are on a computer and trying to access a lesson or module resource for which you have not yet completed the prerequisites, it will look like the screenshot below. You will see a notification that the module has not yet been unlocked and a list of the items that must be completed before you are able to open it. You must complete all of the Prerequisites before you can open a resource that is housed in a module.

- Home
- Announcements
- Modules
- Meet with Mrs. K
- Access Pearson
- Grades
- Syllabus
- Chat
- People
- Zoom
- NetTutor

Week 2 Overview

This page is part of the module **Week 2 - Functions & Their Graphs; Linear Functions** and hasn't been unlocked yet.

Completion Prerequisites

The following requirements need to be completed before this page will be unlocked:

Beginning of Semester Stuff (BOSS)

[Welcome to College Algebra!](#)

must view the page

[Course Navigation and Module Requirements](#)

must mark as done

[Instructor Information](#)

must mark as done

[Student Profile](#)

must submit the assignment

[Course Grades](#)

must mark as done

[Course Materials](#)

must mark as done

[Course Tools & Resources](#)

must mark as done

[Academic Integrity](#)

must view the page

[Academic Integrity Quiz](#)

must score at least a 3.5

[Credit Hours & Workload](#)

must mark as done

[Here's An Unpopular Opinion](#)

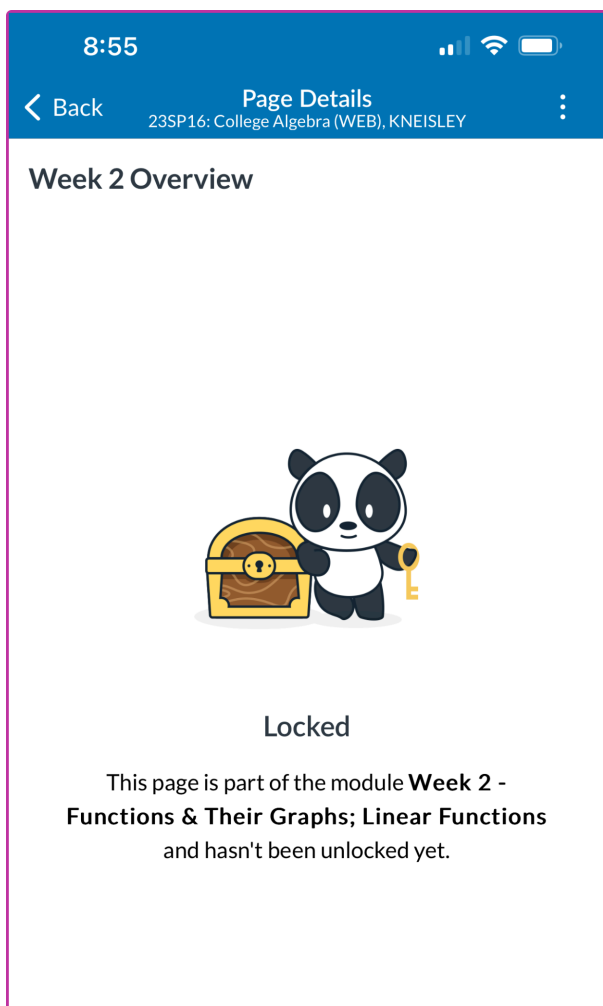
must contribute to the page

[BOSS Module Assignment](#)

must submit the assignment

On a Mobile Device (DO NOT USE THIS OPTION)

If you are trying to access a lesson or module resource for which you have not yet completed the prerequisites, it will look like the screenshot below. You will receive a notification that the item is part of a module that is not yet unlocked. Again, the module or lesson **has** been unlocked by me but you will need to get on a computer to see what you must go back to in order to view the materials.



If you are having any trouble completing the BOSS module items, please contact me ASAP so that I can help you before you fall behind in the course. Contact through Canvas Inbox messages is preferred.



Course Resources & Tools

Everyone gets stuck and struggles with Math sometimes. When this happens, it is important to be aware of the resources that are available to help you get unstuck. Here are a few to get you started.

Resources Available through Midland College

- Dr. Kneisley, your instructor.
- Peer tutoring through the **Midland College Math Lab** (not in *MyLab*) will be held in person in MHAB 125.
 - Please see the [Chap Success Center](#) or email tutoring@midland.edu for more information about availability.

TO DO: Sign in at the Math Lab if you are local to Midland.

Resources Available in this Canvas Course

- Ask a Peer|Help a Peer discussion
 - Research shows that peer-to-peer advice is valuable for learning, so we have created this forum, "Ask A Peer|Help A Peer." This is a place to ask your peers questions and share your knowledge relating to the course, topics, readings, and assignments.
- Tutor.com
 - This online tutoring platform is available to all Midland College students and can be accessed via the left-hand navigation menu in your Canvas course.

TO DO: [Subscribe](#) to the Ask a Peer|Help a Peer discussion board.

TO DO: Access Tutor.com.

Reliable Resources Available on the Internet

Writing in Math

[Tips for Writing Math Well](#) from the Harvey Mudd College Department of Mathematics.

Online Tutorials

The following list includes creators with extensive collections of videos on a wide range of topics. While they may not complete a problem the same way Dr. Kneisley would, what they do is typically accurate and if it helps you understand, great!

- [Khan Academy](#)
- [Math Vids](#)
- [Patrick JMT](#)
- [Purple Math](#)
- [Prof Rob Bob](#)
- [Paul's Online Notes](#)
- [Math and Science](#)

TO DO: Log in to your student YouTube account (or the one you use most) and subscribe to one or two content creators from the list above. You could also add a creator to the list by [making a comment](#).



Math Anxiety

Math Anxiety

The American Psychology Association defines anxiety as follows:

An emotion characterized by [apprehension](#) and [somatic](#) symptoms of [tension](#) in which an individual anticipates impending danger, catastrophe, or misfortune. The body often mobilizes itself to meet the perceived threat: Muscles become tense, breathing is faster, and the heart beats more rapidly.

Anxiety may be distinguished from fear both conceptually and [physiologically](#), although the two terms are often used interchangeably. Anxiety is considered a future-oriented, long-acting response broadly focused on a threat, whereas fear is an appropriate, present-oriented, and short-lived response to a clearly identifiable and specific threat.

When anxiety is directed toward math, it can lead to avoiding the “threat” altogether, which can then start or perpetuate a destructive cycle where avoidance leads to poor performance, poor performance leads to more anxiety, and more anxiety leads to more avoidance.

The good news is that you **can** learn to reduce it overall and manage it when it happens.

Managing Anxiety

It is incredibly important to reduce your overall anxiety because not doing so will ultimately *increase* it. While you are doing so, though, it is almost guaranteed that you will experience anxiety when you encounter your triggers.

There are many research-backed methods for dealing with anxiety in those moments. Please keep in mind that these strategies are typically most effective when practiced. All of these strategies can be done during or right before exams or encountering another anxiety-inducing situation.

- [Reflective Writing](#)
- [Breathing Exercises](#)
- [Coloring Mandalas](#)
 - [Free coloring pages](#)
- [Progressive Muscle Relaxation](#)
- [Anchoring](#)

Reducing Anxiety

Even if you are not dealing with anxiety that is preventing you from performing daily or required tasks or participating in desired activities, you could benefit from therapy if you experience anxiety that impacts you in a negative way. Please see [these resources](#) available to MC Students.

If you do not wish to or are unable to pursue a therapeutic solution or if you need help while you wait for an appointment, please [see this workbook](#) from the University of Michigan for developing a plan for reducing your anxiety.

Another way of reducing your anxiety toward math is to [recondition the response](#). You will need to spend some time practicing this everyday.



Rewiring Your Brain

Rewiring Math Anxiety

A Guide to Using Relaxation and Repetition to Feel Better About Math

Why This Works

Your brain is great at learning patterns. Right now, it might associate math with *stress*. But with practice, you can teach yourself to connect math with *calm & control*. This process combines classical conditioning, systematic desensitization, and positive reinforcement to recognize a new pattern.

First, you will train your brain with a specific relaxation technique, called anchoring, to achieve a relaxed feeling in response to a phrase of your choosing. Then, you will imagine and actually put yourself in increasingly stressful scenarios while you use your anchor to induce calm. Finally, you will reward yourself when you have moved forward.

Step 1: Notice What You're Feeling

Find a quiet place to sit or lie down. Take a deep breath and check in with your body:

- Are your shoulders tight?
- Is your heart beating fast?
- Are your thoughts swirling?

Just observe. No need to fix anything yet.

Step 2: Imagine a Safe, Relaxing Place

Close your eyes and picture the most relaxing place you can imagine. This can be real or imagined. Ask yourself:

- What do I see?
- What do I hear?
- What does the air smell like?
- What do I feel on my skin?
- What tastes are there, if any?

Example: A porch swing in the mountains, light rain, pine trees, holding warm coffee...

Do this daily for a few minutes. Notice how your body responds—slower heartbeat? Looser muscles? You're making progress.

Step 3: Choose Your Phrase

Pick a short phrase to help you connect with this calm feeling. Any of these will work, but choose something personal, memorable, and natural for you:

- "I am safe."
- "Flurm the blibble blop."
- "I'm a chill potato."
- "Calm lives here."

Repeat your phrase slowly while imagining your relaxing place. Do this daily until just saying the phrase starts to calm you.

Step 4: Use the Phrase by Itself

Practice saying your phrase without the imagery. Sit quietly, say the phrase, and observe your body. If you don't feel relaxed yet, go back to using both the image and phrase together for a few more days. You're training your brain which takes time and kindness.

Step 5: Add Math (Slowly!)

Now that your calming phrase works, gently practice using it in math-related situations. Start small and work your way up. If a situation is hard when you are actually doing, try imagining it first while using your anchor, then

Start Here – Low-Stress Situations

Check off when you *consistently* feel calm doing these with your anchor phrase:

- ☐ Opening my math book
- ☐ Looking at a review problem I already understand
- ☐ Skimming old class notes

Next Steps – Mild to Moderate Stress

Practice your phrase here until anxiety fades:

- ☐ Doing a few problems from a familiar homework set
- ☐ Watching a math video or review lesson
- ☐ Reading through a new concept without solving anything
- ☐ Trying a few problems with examples nearby

Eventually Try – Higher-Stress Situations

Use your phrase, go slow, and repeat until you feel confident:

- ☐ Starting a full homework set on a new topic
- ☐ Working through a challenging new concept
- ☐ Studying for a quiz or exam
- ☐ Taking a practice test

- ☐ Going to tutoring or asking for help
- ☐ Attending a math class where I feel behind
- ☐ Taking an actual math test or exam

☒ Every time you check off a box, you're building calm and confidence.
 Celebrate progress, not perfection.

Step 6: Reward Yourself

After each success (even small ones!), reward yourself:

- A snack
A playlist break
- A walk or stretch
- A kind word to yourself ("I showed up today")

This teaches your brain that math is not just safe, but satisfying and rewarding.

Notes & Reminders

- It's normal to have setbacks.
- Go back a step anytime you need.
- Practice daily. Even 3–5 minutes helps.
- You are not "bad at math"; your body just needs to feel safe again.
- You didn't develop math anxiety in a few days. It has likely been reinforced over and over for a decade or more, so it will take much longer than a few days to overcome it.

You're not just learning math. You're learning emotional regulation and self-trust.

Math Study Skills

Math Study Skills

[The Learning Scientists](#) are a group of educators and researchers who aim to provide research-based best practices for studying and learning. Their focus is on 6 major study strategies that can be used for ANY subject.

When You Study

The first two strategies are used to determine **when** you study material.

Spaced Practice

Think about a young baseball player who is learning to hit fastballs. During batting practice each day, the pitcher is going to throw a bunch of fastballs so that the player can work on his timing, the point of contact, and recognizing what a fastball looks like coming out of the pitcher's hand until he has it down and is *good* at it.

This is [spaced practice](#), or spreading out the work at regular intervals, which gives your brain time to develop the neuron pathways to deal with the ideas and concepts.

Interleaving

If batting practice stops here, then the player has not learned how to hit a fastball when it counts. While he might get *really good* at hitting fastballs, an opposing pitcher in a game is not likely to throw him another one once he realizes that the player has done the work to hit a fastball. Instead, the practice pitcher also needs to spend some time throwing a variety of pitches so that the player can practice seeing fastballs out of context, learn to spot the differences between how a fastball comes at him as opposed to how a curveball or change up looks, and adjust his swing accordingly.

This is [interleaving](#), or switching between topics and ideas to help you learn to distinguish between concepts and when to apply different strategies.

Incorporating Spaced Practice & Interleaving

It is almost impossible to “accidentally” use spaced practice or interleaving. To ensure that you are using these strategies, you must make a plan to do SOMETHING *everyday*. Even five minutes a day can be effective if it is done *consistently*. Use one of these resources to develop a plan that will help you space your studying out and use interleaving.

- [Create a weekly schedule and a semester plan](#) (use your syllabi for this!)
- [Five-Day Study Plan for Exams](#)

How You Study

The remaining strategies are techniques used to do the actual studying and or/learning.

Retrieval Practice

Think about going to dinner at a new friend's home. Imagine they ask you to set the table while they step out to take a phone call. You are in a pickle because, since you have never been there before, you don't know where anything is. You just have to start opening cabinets and drawers until you find plates, cups, dinnerware, and salt and pepper. Eventually, you will find everything you need and get it on the table.

Now imagine that dinner went well, and you and this person are starting to become good friends. They invite you over for dinner again the next week and once again ask you to set the table. This time you may still have to open some unnecessary cabinets, but fewer than last time and you are able to set the table much faster.

When you have gotten out the stuff you need to set the table over and over again, you eventually get to the point where you know where the stuff is and don't have to think about which cabinet to get a cup from or where the dinner knives are. You can go directly to the spot where those items are stored immediately and without effort.

This is [retrieval practice](#), or recalling previously learned concepts and procedures without using notes or resources, and is actually **the most important strategy** and you should do this *every single time* you study. But it is *hard* and can lead to being under-confident in your abilities compared to just rereading your notes or simply working old homework problems with your notes in front of you. If you have ever experienced "going blank" on a test after you felt confident and well-prepared, a lack of properly done retrieval practice might be the cause.

Here are some ways you can incorporate retrieval practice:

- Flash cards - most people do not use flashcards in a way that is actually beneficial. If you feel like they are a waste of time, take a look at these three resources for using flashcards and give them a try. You may find that the way you use them matters more than you thought.
 - [How to Study with Flashcards](#)
 - If you get stuck with the instruction cards, use ChatGPT or another generative AI program to help generate some.
 - [How To Use Flashcards Effectively](#)
 - [The 5-Sided Flashcard](#)
- [Take a Timeout](#)
- [Quizzing/Self-Testing](#)

Other Techniques

These techniques are effective strategies on their own but can be used as part of your retrieval practice.

- [Elaboration](#)
- [Concrete Examples](#)
- [Dual Coding](#)

Resources for Studying for Math

What resources you use to prepare to communicate your understanding is up to you. Those resources could include but are not limited to the following. Please keep in mind that, as described above, studying effectively and efficiently for math includes far more than doing the assigned homework problems.

- eBook textbook
- Physical copy of the book
- Problems from the book
- Additional text books
- Free worksheets with answer keys found online
- Reviews (in Canvas)
- Lecture Prep for each section (worksheet available in Canvas for use in conjunction with the book)
- Effective study strategies as described above

An Often Overlooked & Underutilized Resource

Students taking online courses often say they feel like an island, alone in an ocean of unknown content with no one else around. Research shows that those who feel connected to their classmates do better in online classes than those who do not. One way to combat this isolation is to create an [*effective*](#) study group, even if it is via Teams or Zoom, and get together regularly to work on course materials. To find a group, post some general times when you are available in the Ask a Peer|Help a Peer discussion or respond to someone who has already posted.



*Credit Hours & Workload

One of the main differences between high school courses and college courses is that your college teacher will cover about the same amount of material in about 30% of the face-to-face time, and, in an online course, like this one, 0% of the face-to-face time. Many students do not make the connection that much of learning in college, in any course format, is done outside of class.

In a regular 16-week semester, three credit hours represent no less than three contact “hours” of classroom or direct faculty instruction and **a minimum of six hours of out-of-class student work each week for fifteen weeks of instruction**, for a total of at least 135 hours (9 hours per week) of instruction and student work. Students in online math courses should anticipate spending an average of **3 hours per week** engaging with the materials provided by the instructor **and 6 hours per week** beyond that working on class assignments and studying.

TO DO: Make a plan for spending 9 hours per week working on this course.

One way to do this is to make a schedule and post it. An Excel schedule can be found [here](#). Feel free to *make a copy of it* and fill it in with all the things you must do during the week. This should include all the trivial things that you have to do (like travel to work and school, bathing, sleeping, and eating), the big things you must do (classes, work, take care of kids), and some downtime every day to do what you want. When you do this, you can find chunks of time that you otherwise would not know you had.

Attendance

Since this is an online class, there are no class meetings to attend. You must, however, complete an academic assignment prior to the end of Census Day (see the [MC Academic Calendar](#) for this date). If you get behind with the course assignments, please do let me know so that we can work out your next steps.



Your Rights

Your Rights as a Student

When it comes to your grades you have two significant rights: the right to privacy & the right to be assessed fairly.

Privacy (FERPA)

Once you enter college, regardless of age (***including minors***), you have the right to privacy. This means that I cannot discuss your grades, or any aspect of your performance, work habits, attendance, participation, or anything else with anyone besides you who does not have an academic need to know (an example of a person with an academic need to know is an advisor who is working with you to register for the next semester). I take this right **very** seriously, thus it is my policy, and that of Midland College, that if you wish me to discuss any aspect of this class that pertains specifically to you, with anyone, you will need to complete a FERPA waiver through the portal and specify what I am allowed to discuss with whom. Further, it is my policy that I will not participate in any discussion until I have verified with the registrar and records office that the FERPA waiver is complete and active. I also will not participate in the discussion without you taking an active role in the conversation.

Complete a FERPA Waiver

- Sign into the portal
- Click on the *Students* tab
- Click on *Transcripts and Forms*
- Click on *FERPA Permissions* (bottom of the page)
- Fill out & submit the electronic form

Fair Assessment

You have the right to be assessed and otherwise treated fairly. Concerns should be expressed as soon as possible to allow for early resolution. Midland College, and I, encourage you to discuss any concerns (grades or otherwise) with me first. If you feel uncomfortable doing so, or if you are not happy with what I tell you, you may contact the Math Department Chair, Dr. Cohlmiia, 432-685-4541 or kcohlmiia@midland.edu, then the Dean of the Math & Science Division, Dr. Poage, 432-685-4561 or mpoage@midland.edu. If a resolution is still not possible, you may proceed with the formal [complaint process](#).



*Grades & Assignments

You will be graded in this course based on your **communicated** understanding of the course outcomes. This communication is most often done by showing your work, but can sometimes be done via written or verbal communication.

Course Categories

This is a points-based course which means your final course grade is determined by adding up the number of points you have earned on all assignments in the course. At the end of the semester, your numeric grade will be determined by dividing the number of points you have by 1000. Points are available in the following categories:

- 350 points from the Midterm Exam
- 350 points from the Final Exam
- 300 to 360 points from other assignments

While there are up to 1060 points available in the course, at the end of the semester, your point total will be divided by 1000, not 1060 or whatever the actual total turns out to be. This means that there is **already extra credit** built into the course. But this also means the **percentage** grade that Canvas shows will likely be lower than what your letter grade will turn out to be, particularly as the end of the semester nears.

Letter grades will be assigned as follows:

Letter Grade	Point Range
A	900 or more
B	800 to 899
C	700 to 799
D	600 to 699
F	0-599

Important Things to Note

- Final numeric grades are rounded to the nearest point using traditional rounding rules before letter grades are assigned.
- Any emails asking me about ways to improve your grade or to give you a letter grade above that which you earned after you have taken your final exam will **not** receive a response. I am, however, happy to discuss any errors you feel that I may have made entering your grades at that point.

Late Work

In an effort to minimize stress and the need for extension requests, I have built in penalty-free late windows for assignments where feasible. The first "soft" deadline (similar to a "best by" date at the food) is called the **Due Date** within Canvas, and this is the date that appears on your Canvas calendar and to-do list. Assignments completed after the Due Date may be updated to LATE automatically by Canvas and may still earn full credit with no late penalty, they may show as a zero until you submit them.

The "hard" deadline (similar to an "expiration" date on food) is the date after which the assignment will become unavailable for further submissions, shown in the assignment details after the word **Available** in Canvas. Assignments not submitted by this deadline **will not** be reopened for submission barring extreme circumstances and lots of communication.

Note: Many Canvas assignments are graded manually. Manually graded LATE assignments will continue to show a grade of 0 until I have completed grading all submissions.

In short: if the assignment lets you in, you may still complete it for up to full credit — even if Canvas says it's MISSING or LATE and your grade shows as 0 for now. ***Do your best to stick to the Due Dates*** and not let things pile up, but you can usually take some extra time without needing to request an extension or worrying about any late penalties.



Course Outcomes

At the end of the course, you should be able to perform each of the following skills.

Outcome Group 1. Equations & Systems

- 1) EQ-1. Quadratic Equations
 - a) I can find all real and complex solutions to quadratic equations using a variety of methods, including factoring, completing the square, and using the quadratic formula.
 - b) I can describe the discriminant of a quadratic equation and use it to determine its number and character of the solutions.
- 2) EQ-2. Other Types of Equations
 - a) I can solve radical equations and check for extraneous solutions.
 - b) I can use factoring to solve equations.
 - c) I can solve equations in quadratic form.
 - d) I can solve rational equations and check for extraneous solutions.
- 3) EQ-3. Exponential & Logarithmic Equations
 - a) I can solve exponential equations.
 - b) I can solve logarithmic equations and check for extraneous solutions.
- 4) EQ-4. Systems of Equations
 - a) I can solve systems of linear equations in two and three variables by substitution and I can solve systems of linear equations in two variables by elimination.
 - b) I can identify inconsistent systems of equations in two variables and I can express the solution of a system of dependent equations containing two variables.
 - c) I can solve systems of linear equations in two and three variables using matrices.

Outcome Group 2. Functions

- 1) FN-1. Analyzing Functions
 - a) I can determine whether a relation represents a function and find the value of a function.
 - b) I can find and simplify the difference quotient of a function.
 - c) I can combine functions using addition, subtraction, multiplication, and division.
 - d) I can identify functions that are even, odd, or neither even nor odd from a graph and an equation.
- 2) FN-2. Piecewise-Defined Functions
 - a) I can evaluate piecewise-defined functions.
 - b) I can find the domain of a piecewise-defined function.
- 3) FN-3. Linear Functions
 - a) I can find the intercepts and slope of a linear function.
 - b) I can use the average rate of change to identify linear functions.
 - c) I can build linear models from verbal descriptions and use linear models to establish conclusions.
- 4) FN-4. Composite Functions
 - a) I can combine functions using composition.
 - b) I can find the domain of a composite function.
 - c) I can find the components of a composite function.
- 5) FN-5. Inverse Functions
 - a) I can determine whether a function is one-to-one.

- b) I can determine the inverse of a function defined by a map or a set of ordered pairs.
 - c) I can find the inverse of a function defined by an equation.
- 6) FN-6. Quadratic Functions
 - a) Given a quadratic equation, I can identify the vertex and the axis of symmetry on its graph.
 - b) I can use an equation or a graph to find the minimum or maximum value of a quadratic function.
 - c) I can build quadratic models from verbal descriptions and use quadratic models to establish conclusions.
- 7) FN-7. Polynomial Functions
 - a) I can identify polynomial functions and their degree.
 - b) I can analyze the graph of polynomial functions.
 - c) I can identify the zeros of polynomial functions and their multiplicity.
- 8) FN-8. Rational Functions
 - a) I can find the domain of a rational function.
 - b) I can find the vertical asymptotes of a rational function.
 - c) I can find the horizontal asymptotes of a rational function.
 - d) I can find the oblique asymptotes of a rational function.
 - e) I can find any intercepts appearing on the graph of a rational function.
 - f) I can analyze the graph of a rational function.
- 9) FN-9. Exponential Functions
 - a) I can evaluate exponential functions.
 - b) I can solve problems involving exponential growth and decay.
- 10) FN-10. Logarithmic Functions
 - a) I can change an exponential equation to a logarithmic equation.
 - b) I can rewrite logarithmic expressions using properties of logarithms.
 - c) I can evaluate logarithmic expressions without using a calculator.
 - d) I can determine the domain of a logarithmic function.

Outcome Group 3. Graphing

- 1) GR-1. Analyzing Graphs
 - a) I can identify local/relative and absolute extrema of a function.
 - b) I can find intercepts from a graph and an equation.
 - c) I can determine where a graph is increasing, decreasing, and constant.
 - d) I can test an equation for symmetry with respect to the x-axis, the y-axis, and the origin (even/odd/neither).
 - e) I can identify the graph of a function and I can use a function's graph to obtain information about the function.
 - f) I can identify symmetry from a graph.
 - g) I can find the domain and range of a function (defined by an equation and/or graph.)
- 2) GR-2. Creating Graphs (*Tested on all 4 unit exams*)
 - a) I can use transformations to graph equations.
 - b) I can complete a graph so that it has a given type of symmetry.
 - c) I can obtain the graph of the inverse function from the graph of the function.
 - d) I can graph piecewise-defined functions

- e) I can accurately graph each of the common functions in algebra, describing any intercepts or symmetry.
- f) I can graph a linear function.
- g) I can graph quadratic functions.
- h) I can graph polynomial functions.
- i) I can graph rational functions.
- j) I can graph exponential functions.
- k) I can graph logarithmic functions.



*Exams

You will take two exams for this course, a Midterm Exam and a Final Exam. The exams are ***paper-based exams*** (not taken on computer) and ***not*** multiple-choice. They are graded with a focus on the work that you show to arrive at your final answer, thus an incorrect final answer with substantially correct work can, in some circumstances, receive full credit while a correct answer with little to no work, or largely incorrect work can and, more often than not, will receive little to no credit. When you take your exam, you may take your own writing implements (pencils, colored pencils, highlighters) and your own [calculator](#), but you are not allowed any [additional resources](#).

While exams may be taken early with enough notice, no exam can be taken late. ***These exams must be taken at an approved in-person testing center and administered in-person by an approved proctor.*** No option to do otherwise exists for this course.

I will collect information from you regarding your plans to take these exams twice this semester via the Midterm Exam Plans and Final Exam Selection & Plans surveys.

Pre-approved locations include the Midland College [Exam Center](#) in the FLRC and the WRTTC in Fort Stockton. Alternate locations must be approved by Dr. Kneisley well in advance of the date that the exam window opens. After you submit the forms for your alternate location and the approval process has been completed, you will receive a message ***to your MC student email account***, informing you of the decision. If you wait until the final deadline to submit your forms and your testing location is denied, you will be required to take it at the Midland College Testing Center in the FLRC. You and any proctor from a facility that has not been pre-approved will need to complete the [Signature Page](#) ***by hand*** before your plans are approved.

Things to Know About the Exam Format & Rules

- Must have a photo ID with you (MC Student ID is acceptable)
- Must be taken in person at an approved facility
- Questions are open ended and not multiple choice in nearly every case
- May use a scientific calculator of your choice (not a graphing calculator)
- May not use additional resources
- May not use additional scratch paper
- Exam must be completed in two hours

TO DO: Complete the Exam Proctoring Plans survey in Canvas.

Midterm Exam

Your midterm exam is worth 350 points and will test you on material from lessons 1-9 (General Functions, Equations, & Quadratic Functions). You will not get a copy of your exam back from me, but I will fill out the rubric attached to the assignment to give you some indication of where your

strengths and weaknesses lie. If you would like to go over it, I am happy to schedule a time with you to do so.

The Midterm Exam Module will not open until you have completed Lesson 9 but a list of skills that are eligible to be tested on the Midterm Exam can be found now on the [Midterm Skills Checklist](#).

See the [Assignment Summary](#) for the exam window.

Final Exam

There are two options for your final exam.

- The first option is worth 350 points and is not cumulative, covering only lessons 10-24.
- The second option is a cumulative test that covers Lessons 1-24.
 - Your score on this exam will be entered as your grade for the Final Exam **and** your grade on the Midterm Exam will be replaced with the equivalent percentage. ***This score will replace the score on the Midterm regardless of whether the score is better or not.***

You will need to let me know which exam you want to take when you submit the ***Final Exam Selection & Proctoring Plans survey***, which does not open until the midterm exam closes.

The Final Exam Module will not open until you have completed Lesson 24 but a list of skills that are eligible to be tested on the Final Exam can be found now on the [Non-comprehensive Final Exam Skills Checklist](#) or on the [Comprehensive Final Exam Skills Checklist](#).

See the [Assignment Summary](#) for the exam window.



No Open Resource Exams

Every semester, students express a strong preference for the use of notes on exams, citing the use of resources in real-world professional scenarios, so I would like to take the opportunity to address that here. I appreciate your thoughts about this and I assure you that the math department has taken the issue into great consideration before arriving at our policy.

I'll begin with an example of a real-world "test." My husband and I built a house in 2019. It turned out that our builder took some significant shortcuts, one of which was the way they attached the stone to the facade of our home. When they refused to fix the wall on our house that was having issues for the second time on the grounds that that was either a maintenance issue or normal wear and tear, we had to hire an engineer to encourage their warranty company to get involved.

The engineer we interviewed knew what to look for, knew how a foundation was supposed to be laid when stone or brick was to be attached to the facade (it's called a "brick ledge" by the way) especially when it is on a hill, as ours is and what it meant that our stone was cracking and moveable. Now, perhaps this was not information that he carried around in his head regularly, but when he came to meet with us, he was 100% prepared to answer nearly every question we had. We hired him to write an official report and, after the warranty company's engineer examined our report, the builder redid the stone on our home. The meeting would have gone very differently had he answered every question with "let me look that up" and I absolutely would not have hired him because I would not be convinced that he knew what he was doing or that he would be able to convince anyone else either.

I acknowledge that if I hire an engineer to do a job, I need him to use all of the tools at his disposal. However, I also expect him to have significant skill with the basic tools, knowledge, and language of the trade and to be able to communicate that to me clearly.

Similarly, as a math student in a gateway course which is one of the first of, what will be for many of you, many math courses that build on this one, you are expected to be able to converse in the basic terminology of the field and to have significant facility with the fundamental knowledge and skills of the mathematical topics at hand. At the end of this course, if you receive a passing score, especially an A or B, it means you are ready to work with and understand the concepts taught in the next course and are in a position to be successful in doing so. Whether that is something you will need or choose to do is not something for which I can adjust the course requirements.

The research on open-resource exams has primarily examined one of two aspects: either student *perception* or student *performance*. As I mentioned above, just as the research shows, a great number of you *perceive* that open-resource exams are better and indeed, they have been shown to reduce anxiety for many high-anxiety students. However, the results regarding *performance* are mixed, with many studies showing that students perform about as well on both open-note and closed-note exams. Beyond this, [one study](#) found that there were significant issues with the notes that students created to use on open-note exams that actually *harmed* them.

Because of the unclear evidence that open-resource exams improve performance and because there are other, more effective ways to reduce anxiety besides allowing the use of potentially inaccurate resources (see [this page](#) or visit with me if that is something you feel like you need), the math department has decided that we will **not** allow open-resource exams on any midterm or final exam in the STEM pathway.

MyLab Assignments

MyLab

MyMath Lab is an interactive online computer software package that accompanies the text or can be purchased without a textbook. *MyMath Lab* is required in this course. Make sure you purchase access to this program, either by itself through the Midland College bookstore or directly through Pearson (because it comes with the book and solutions online) or with a textbook. If you buy a physical book, be sure the book has *MyMath Lab* Access. USED books and eBooks do NOT include *MyMath Lab* Access. *MyMath Lab* Access may NOT be included with a NEW book purchased from an online retailer such as Amazon.

What is Included in MyMath Lab?

- Online access to all pages of the textbook, including solutions
- Online homework problems like those in the textbook with instant feedback
- Multimedia learning aids (videos & animations) for selected examples and exercises in the textbook
- Personalized study plan based on performance on online exercises and quizzes
- Skill Builder provides timely and adaptive support for individual problems

For this Course

The final submission deadline for all of the assignments described below is 11:59 PM on the deadline listed in the [assignment summary](#).

Assignments are arranged in MyLab so that you must attempt the Ungraded Practice for a lesson and score at least 10% prior to attempting the Graded assignment for that same lesson. You also must score at least 50% on the Graded Assignments prior to opening the Review Quizzes for the exams. The Canvas modules are arranged similarly. Keep in mind that while navigating through Canvas, Pearson does not sync your grades instantaneously. It may take a few hours for it to update, so waiting until Sunday to work through the module is NOT a good idea.

Consider these scaffolded assignments as the practice activities of a theater performer. The Ungraded Practices are like the practice they do at home: running lines, vocal exercises, practicing choreography, perhaps even cardio exercises. The Graded Assignments are like the rehearsals with the cast where they work on particular scenes or dances, and have to use what they worked on at home. Finally, the Review Quizzes are like a dress rehearsal in which they run through the entire performance, in costume and on time, without stopping. After the dress rehearsal, the director will have notes on what needs to be tightened up, but the significant majority of the work would have been done prior to this.

Please also note that MyLab is **very particular** about how you enter answers. Many of the answer entry boxes have instructions in blue underneath them to help you with this. If you would like me to review answers that you believe are fundamentally correct but entered incorrectly, the best way to let me know is to **use the Ask My Instructor feature in MyLab**. This will send me a direct link to the

problem and I do not have to guess which ones you mean. Barring this, you are welcome to email me through Canvas or at my college email.

Ungraded Practice

The ungraded practice assignment for each lesson is just that - practice. They are designed to help you prepare for the graded assignments in the course. These assignments provide instant feedback and give you an opportunity to see every type of problem that could be on the Graded Assignments and Review Quizzes. They are untimed and you may attempt them as many times as you like. You are welcome to use all of the resources you have at your disposal, including your notes, the resources available in MyLab, symbolic AI like Wolfram Alpha or Symbolab, or calculators such as Desmos. You may do as much or as little of these assignments as you like, but you must score at least 10% to move on to the next item in the module.

Graded Assignments

Each lesson is accompanied by a graded assignment that consists of a much smaller selection of problems than the ungraded practice and [has a few review problems at the end](#).

These assignments are untimed and you have unlimited attempts, but the MyLab resources are turned off and you should not use any resources other than your notes. You will also not receive feedback on each problem until the end of the assignments. All passwords for these assignments are on the assignment page in Canvas. Please see the [Course Navigation](#) section about accessing these pages.

Review Quizzes

There are two graded Review Quizzes, one for each of the Midterm and Final Exams. The quizzes are timed, but you have unlimited attempts. They are intended to help you determine where you stand, what you need to prioritize before taking the exam, and practice pacing yourself.

You are expected to complete these assignments on your own without help from other people or physical (e.g., notes, textbooks) or online sources (e.g., Khan Academy, symbolic AI like Wolfram Alpha, graphers like Desmos).

Consider these scaffolded assignments as the practice activities of a theater performer. The Ungraded Practices are like the practice they do at home: running lines, vocal exercises, practicing choreography, perhaps even running or doing other workouts. The Graded Assignments are like the rehearsals with the cast where they work on particular scenes or dances, and have to use what they worked on at home. Finally, the Review Quizzes are like a dress rehearsal in which they run through the entire performance, in costume and on time, without stopping. After the dress rehearsal, the director will have notes on what needs to be tightened up, but the significant majority of the work would have been done prior to this.

Writing Assignments

Participating in discussion boards and self-reflection assignments in this course is crucial as it fosters mathematical reasoning, enhances interdisciplinary understanding, and promotes metacognitive skills development. Through writing tasks, you can internalize effective communication strategies and deepen your learning experiences, while self-reflection enables you to identify learning gaps, tailor study strategies, and take ownership of your academic progress, ultimately leading to improved performance and engagement in mathematical concepts and beyond.

Check-Ins

Five times throughout the semester, you are asked to check in with me to let me know how things are going. These assignments can be completed by writing as little as a sentence or two, but you are welcome to write more. While I will read every response submitted, I will not be able to respond to every one of them. No assignment will be accepted beyond the date that the assignment closes.

- Check-In 1 (Due Sunday of Week 3)
- Check-In 2 (Due Sunday of Week 6)
- Check-In 3 (Due Sunday of Week 9)
- Check-In 4 (Due Sunday of Week 12)
- Check-In 5 (Due Sunday of Week 15)

Discussions (Extra Credit)

There are 2 discussion boards you are responsible for posting to this semester. The first is a low-stakes getting-to-know-you type of thing while the second is open all semester and serves as a place to give and receive help. Each post on the discussion board is expected to follow the [rules of netiquette](#).

- [Introduction Discussion](#)
 - Participation in this discussion is mandatory to move on to the content of the course. Extra credit will only be awarded for posts submitted before the deadline (see the [assignment summary](#)).
- [Ask a Peer | Help a Peer](#)
 - Participation is optional. but highly encouraged. It is open all semester and serves as a place to ask questions and receive answers. Extra credit is awarded for both posting questions **and** answers (up to 10 points).

AI Policy for Writing Assignments

Unless the assignment instructions indicate otherwise, for all assignments described, you are expected to respond using your own thoughts. You are welcome to use generative AI (e.g., Chat-GPT) to edit your responses for clarity, conciseness, or professionalism so long as you credit the source (see [this description](#) of the APA 7 guidelines for in-text parenthetical citations and full citations) but submitting work as your own that was created using generative AI (like Chat-GPT),

symbolic AI (like Wolfram Alpha), or other unapproved sources represents academic misconduct and will be reported as such.



Honors Projects & Research

Research

Midland College offers [Undergraduate Research opportunities in Mathematics](#). We use a process called [mathematical modeling](#) to answer big messy problems such as

- Developing a food plan for a mission to colonize Mars,
- Electrifying the nation's fleet of vehicles,
- Locating medical facilities and allocating resources in rural areas, or
- Predicting need and developing a plan for locating public resources for affordable housing in a given metropolitan area.

In your first semester of research, we will teach you the process and you will complete 3 practice problems. In subsequent semesters, we will work with you to design your own problem based on a need you see in a community you belong to, develop a solution to that problem, and, potentially, communicate that solution to stakeholders with the ability to implement the solution.

Students in their second semester of research and beyond can receive a small [stipend](#) for participation. Additionally, students who complete three or more semesters of research are named MC Research Fellows and receive recognition at graduation and on your transcript.

If you are interested in participating in research, please complete [this form](#) and I will contact you soon.

Honors Projects

All of my [honors projects](#) have three components: a written component (such as a paper, a script, or a document produced in LaTeX), a visual component (such as a painting or a model), and an oral component (typically a presentation during a [poster session](#)). We will work together to determine the details of your project and then work closely throughout the semester to ensure that you are able to complete a quality project.

The ideas listed do not constitute a full list of the possibilities for your project. If you wish to do something else, you are welcome to do so.

If you are interested in doing an honors project for this course, please complete the [Honors Program application](#) and fill out [this form](#). I will contact you soon.

Suggested Project

Participate in MC Math Research. You are allowed to “double dip,” particularly during your first semester when no stipend is involved. Subsequent semesters will be up to your instructor for the current math class.

Video Projects

You should expect to produce between 10 and 25 minutes of content, provide a script or transcript of the video, a bibliography of at least 3 sources per video, and a summary of what material each source provided. These projects may or may not require a bit (or more) of video editing, depending on the project.

- [Mathematical Minutes Videos](#) (You will do 4-6 videos, each 3-5 minutes long)
- [Vi Hart](#) inspired Math Doodles
- [Proofs without words](#)
- Original Skits

Other Projects Ideas

- Hypothetical scenarios
- Historical research involving the development of techniques
- Explorations of additional topics or more advanced techniques
- How-To Guides/Summaries of course topics
- "Cheat Sheets"
- Art in Math

TO DO: If you are interested in participating in Research or an Honors Project, complete the [interest form](#).

Tab 17

